Supplementary Information — X4

Fixed study data and provenance; complete machine-readable source tables are included in this package.

S1. Input provenance and implementation

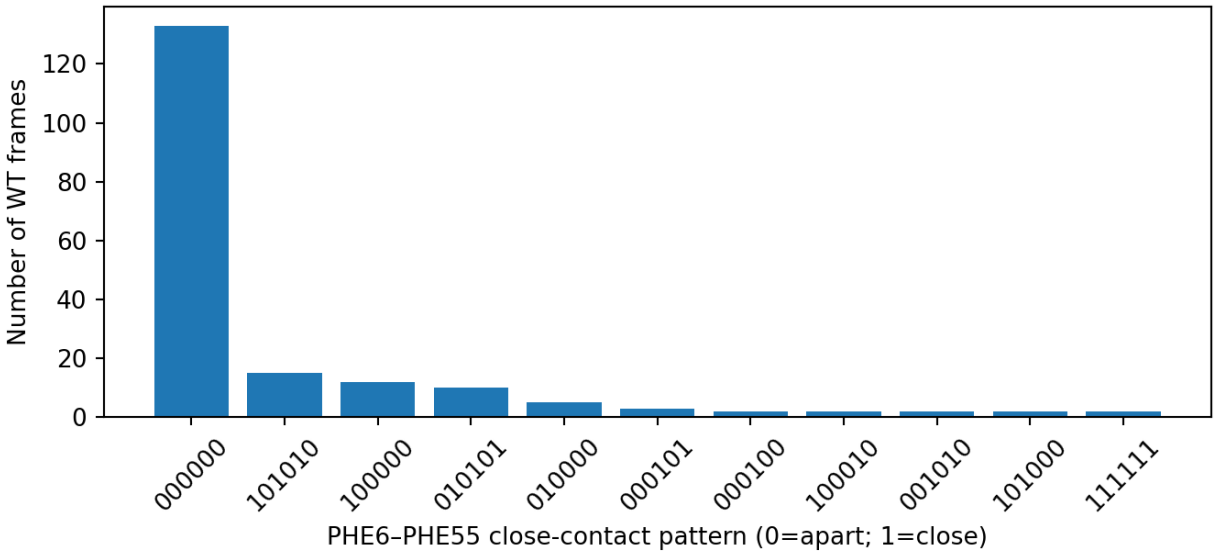
One BioEmu WT ensemble: 200 source frames, six stored HPacker stages (1,200 PDBs). HPacker commit 6cd5ec33961bb2662b51c0a15d11d6fd9162f2a7; subsequent BioEmu commit 42c551133f9825f4ec01812075b0d50e268c2209. Complete original end-to-end NN replay has not been independently established.

Source	Evidence / distributed file
Stage audit	data/all_1200_stages.csv; data/frame_summary_200.csv
Selected v8.1/v8.2 geometry	data/V8_1_STAGE_GEOMETRY.csv; data/V8_2_STAGE_RESULTS.csv
v8.3/v8.4 response	data/TEN_PATH_MODEL_RESPONSE_COMPARISON.csv; supplement/Table_S4_selected_response_evidence.csv
Phase C	Four archived case events and summary JSON files
Stage A	data/stage_a_summary.json; pairwise LJ difference CSVs

S2. Stage-wise close contacts

Strictly <1.6 Å nonadjacent interresidue heavy-atom pairs; separate PHE6–PHE55 criterion. An old general_same_atom_recontact counter did not require an intervening absence and is not a valid recontact count. The PHE-specific ordered recontacts (43/200) do require an intermediate noncontact.

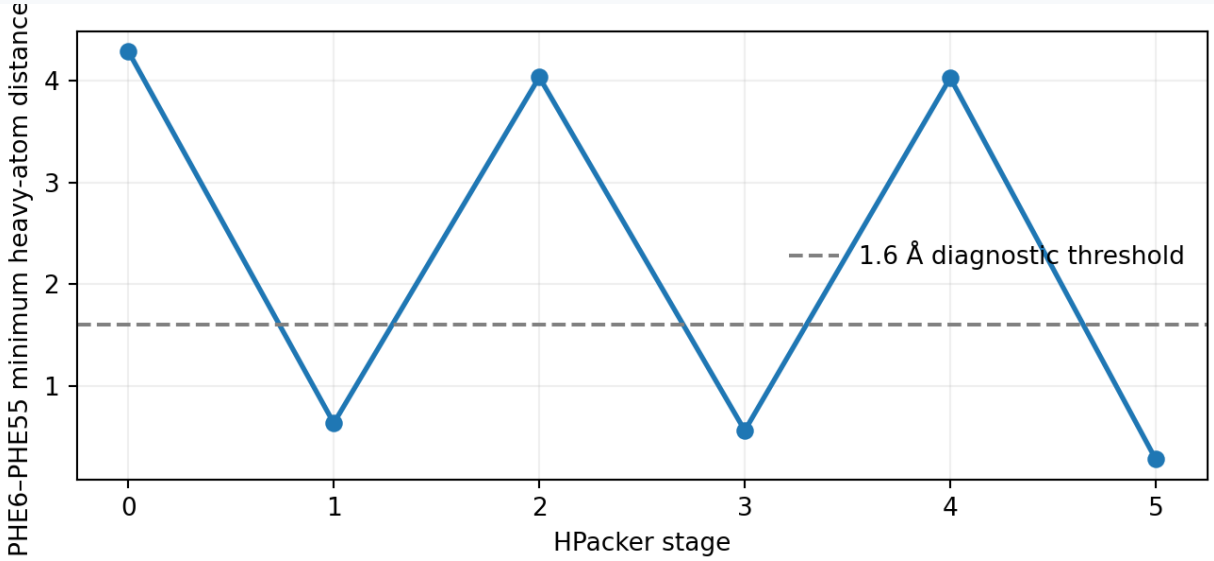
Stage	Any close / 200	PHE6–PHE55 / 200
0	183	39
1	148	22
2	140	25
3	128	23
4	130	29
5	134	21



S3. PHE6–PHE55 selected stage structures

WT0 stage 0–5 PHE6–PHE55 minimum distances: 4.291, 0.633, 4.037, 0.560, 4.030, 0.277 Å. Main-text six-panel display uses stored PDBs; distances are calculated in 3D. Fully alternating contact-pattern frames: 010101 = 0, 10, 34, 53, 73, 83, 88, 167, 180, 191; 101010 = 14, 28, 49, 60, 66, 77, 113, 126, 134, 137, 141, 145, 160, 165, 190.

Pattern	N / 200	Stage 5 state
010101	10	Close
101010	15	Apart



S4. Local perturbation and NN response diagnostics

The v8.1 masking branch removed selected nonbackbone side-chain atoms of the designated residue from the neighborhood arrays res_ids, atom_names, elements, coords, and charges. It did not delete the residue from the original PDB. In the lag branch, the selected side chain was temporarily supplied from stored stage s−2, then restored after input-neighborhood calculation; s < 2 was ineligible. Corrected structural interventions changed the relevant mediator χ prior to rebuilding Cartesian side-chain coordinates (the earlier output-only v8 edit did not). Experiment-specific stage, target, mediator and branch assignments must follow the original archived runner; these general rules alone are not a comprehensive forward-replay certificate.

Stored case	Operation / selected interpretation
WT0 v8.1	Stage 2, target 6, intervened 55; third residue 58; lag branch; mediator 17
WT1 v8.1	Stage 1, target 55, intervened 6; mask branch; mediator 56
WT0 v8.2	Subsequent residue-17-associated intervention and downstream residue-53 investigation
WT1 v8.2	Residue-56-related operation; downstream 55/59/3 responses examined
v8.3/v8.4	Post hoc selected strong/weak comparisons; no prespecified binary threshold

Representative selected maximum circular χ responses were WT0 53→55: 54.089°; WT1 55→56: 178.329°; 55→3: 148.689°; 55→59: 52.082°. Weakly responding selected controls had nonzero raw NN differences. Zernikegram L2 distance, maximum raw8 difference, χ circular distance, Stage and donor/receiver IDs are stored in Table_S4_selected_response_evidence.csv. The WT0 17→55→56 and WT1 branching descriptions remain exploratory proposed response chains, not independently proven unique mediating pathways. Phase 2B is not independently re-audited.

S5. Phase C minimization and short relaxation conditions

Fixed BioEmu run_one_md with only_energy_minimization=False and simtime_ns=0, OpenMM 8.2.0, actual OpenCL platform. System: amber99sb.xml + tip3p.xml, 1.0-nm padding, 0.1 M NaCl, PME 1.0 nm cutoff, hydrogen bond constraints and rigid water. Temperature 300 K and 1 bar pressure in NPT. The source uses LangevinIntegrator, with initial friction 200 ps^{-1} and step sizes 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3} ps, 0.1 ps each. Main 1-fs

timestep; NVT 0.1 ns with 10 ps^{-1} friction, NPT 0.4 ns with 1 ps^{-1} friction and MonteCarloBarostat. Backbone N, CA, C, O are position restrained by CustomExternalForce k*periodicdistance² with k=1000 (as implemented, not an automatically equivalent ½kx² parameter). OpenMM minimizeEnergy precedes integration. OpenMM source-default settings not explicitly overridden in the code are version-dependent. The read-only checkpoints split step calls to ≤1,000 steps and may alter stochastic path timing.

Phase	Time / steps	Conditions
Initial stabilization	0.1 ps each / 111,100 steps	4 increasing timesteps; after minimization
NVT	0.1 ns / 100,000 steps	300 K, 10 ps^{-1} friction; positional restraint
NPT	0.4 ns / 400,000 steps	300 K, 1 bar; 1 ps^{-1}; positional restraint
Production	0 ns	Not executed

Case	Status	Runtime	Final solvated PE (kJ/mol)
WT0 baseline	PASS	32m 46s	−143055.18
WT0 rescue	PASS	33m 30s	−146432.79
WT1 baseline	PASS	33m 16s	−145726.65
WT1 rescue	PASS	33m 18s	−146144.45

Both WT0 structures reached zero <1.6 Å diagnostic close contacts after minimization. Total solvated PE values must not be interpreted as free-energy differences or compared between differently solvated systems.

S6. Stage A heavy-atom Lennard-Jones decomposition

AMBER99SB parameters for the common 494 original protein heavy atoms only; nonadjacent interresidue pairs with appropriate exclusions. No solvent, ions, hydrogen-pair LJ, PME electrostatics or bonded-energy contribution; temporary OXT was used for force-field typing only. The pairwise LJ proxy is not the Phase C total solvated potential energy.

Geometry	Net LJ (kJ/mol)	Pairs <1.6 Å
WT0 baseline	$+1.662\times10^{13}$	6
WT0 rescue	$+2.707\times10^{4}$	1
WT1 baseline	$+9.951\times10^{6}$	3
WT1 rescue	$+2.758\times10^{4}$	1

Largest contributors: WT0 PHE6 CD1–PHE55 CE2, 0.277→5.514 Å, ~99.9999991% of net proxy reduction; WT1 PHE6 CG–PHE55 CE2, 0.916→6.336 Å, ~99.1753%. These are particular atom-pair distances, not necessarily the minimum distance between residues.

S7. Verification, chronology and boundaries

The main archive analysis, selected input/NN diagnostics, MD continuation and LJ proxy are separate evidentiary tiers. WT0/WT1 source selection and strong/weak path labels were exploratory and post hoc. The original X4 core was summarized 21 October 2026; selected continuation stopped 23 October 2026 after Stage A. Full Phase 2B independent re-audit, exhaustive HPacker forward reproduction, Stage B and Stage C decompositions were not performed. Manifest checksums and source records are in provenance/ and SHA256SUMS.txt.